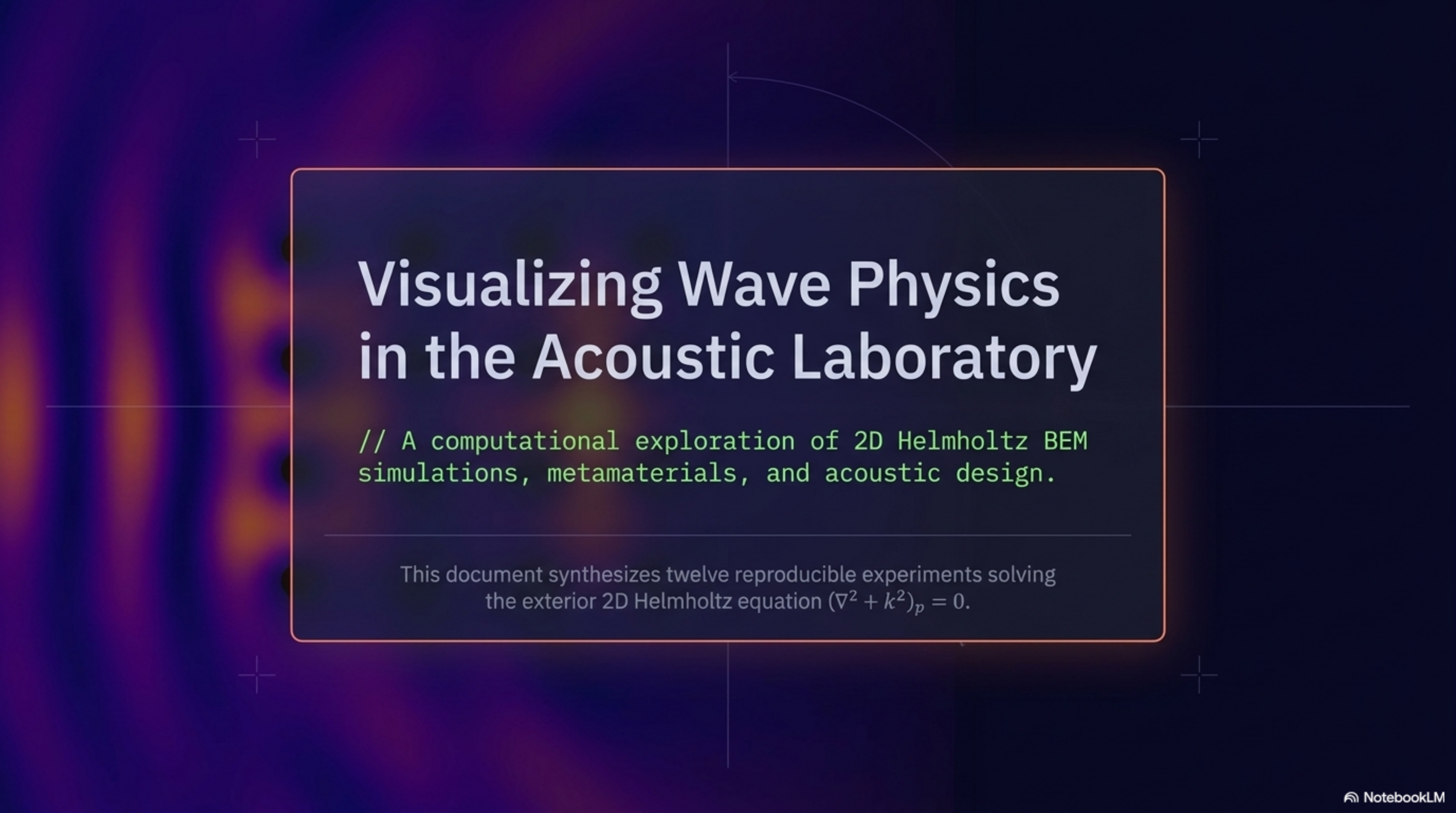




Visualizing Wave Physics in the Acoustic Laboratory

// A computational exploration of 2D Helmholtz BEM
simulations, metamaterials, and acoustic design.

This document synthesizes twelve reproducible experiments solving
the exterior 2D Helmholtz equation $(\nabla^2 + k^2)_p = 0$.

The computational engine powering real-time BEM exploration

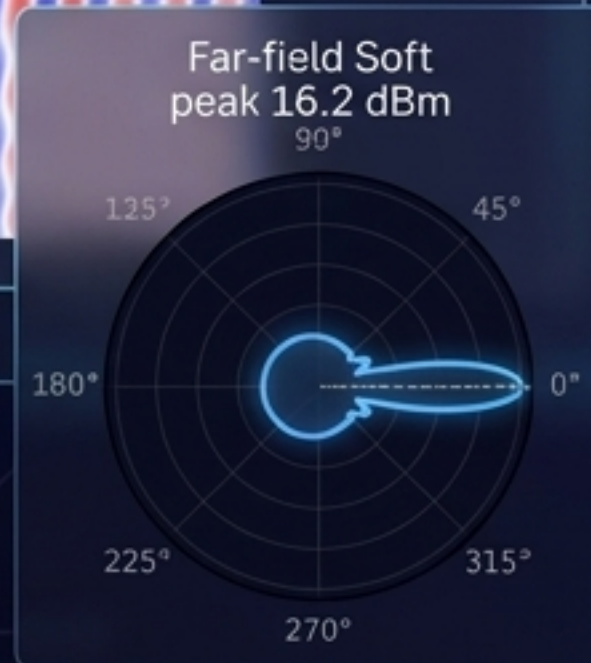
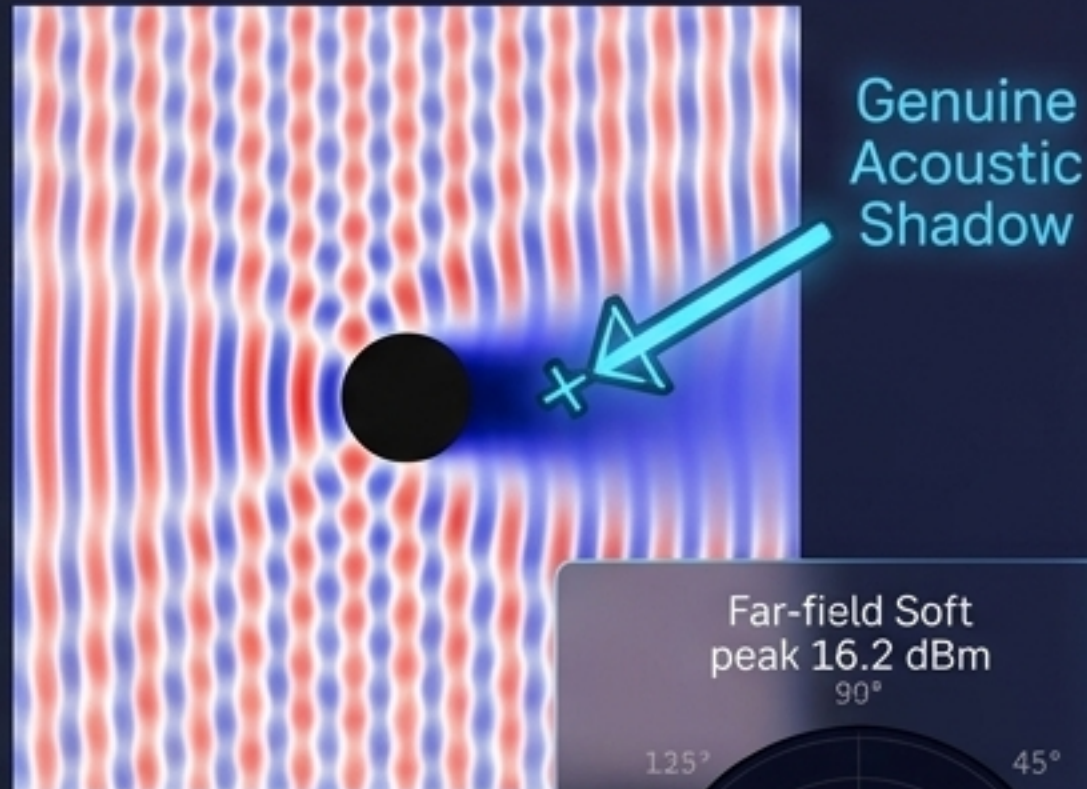
Solving wave scattering requires assembling and solving complex, dense matrices. The new **gpu_bem** backend processes the single-layer BEM representation at blistering speeds, enabling interactive experimentation.



The fundamental material divide governs all acoustic scattering

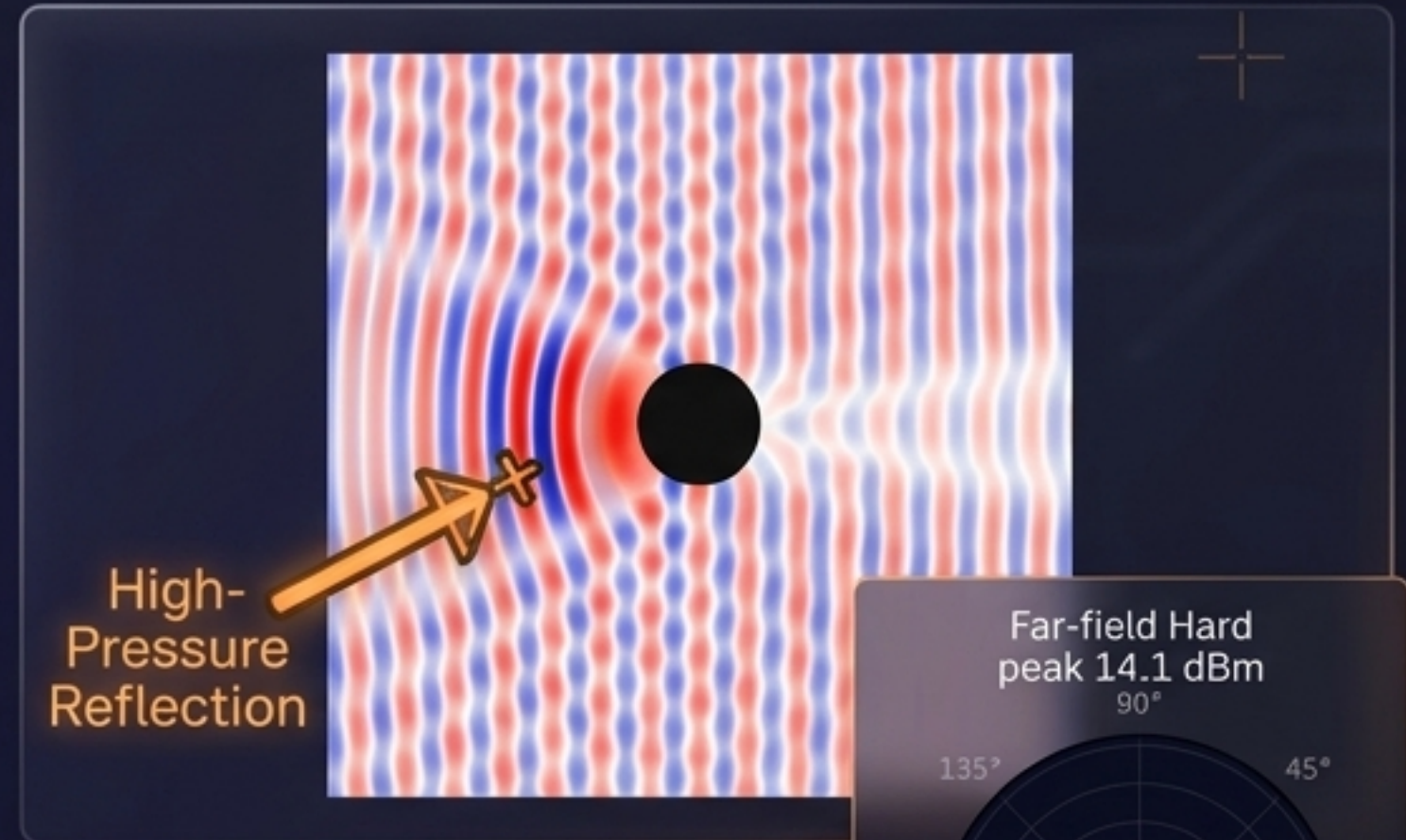
Soft Boundary (Dirichlet): $p = 0$

Pressure-release surfaces (vegetation, foam) absorb and cancel energy, creating genuine acoustic shadows.



Hard Boundary (Neumann): $\partial p / \partial n = 0$

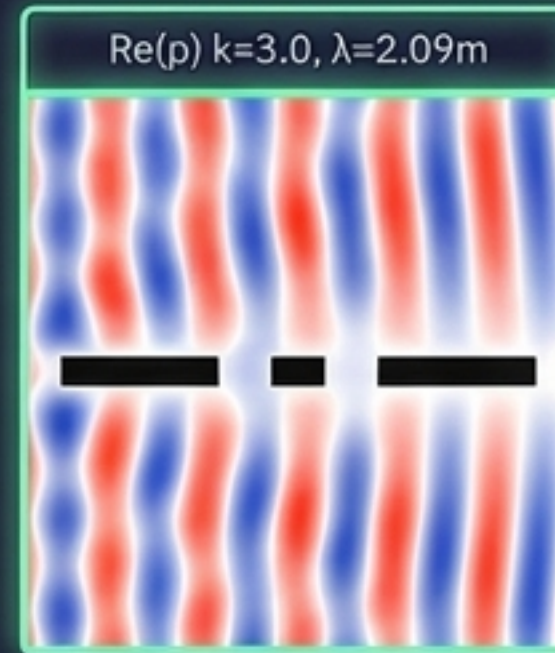
Rigid surfaces (concrete, steel) act as mirrors, preventing energy from leaving the domain and creating bright reflections.



The Helmholtz equation scales universally across wavelengths

Double Slit Dynamics

Plane waves diffracting through two apertures create constructive/destructive interference stripes, identical to optical wave behavior.



$$\Delta y \approx \frac{\lambda L}{s}$$



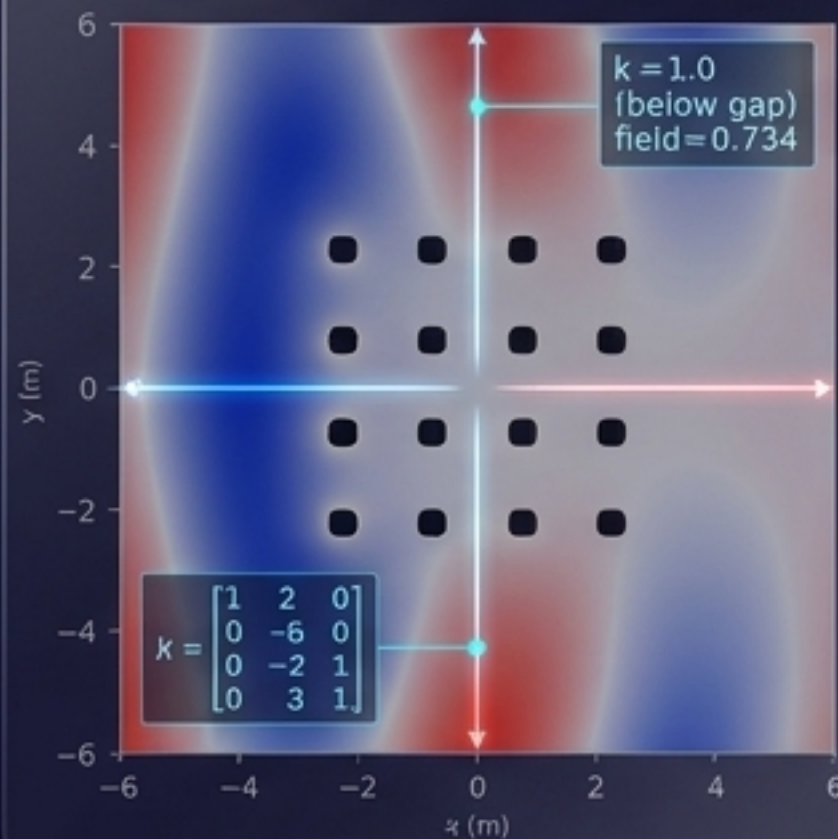
Lloyd's Mirror Effect

A point source interfering with its own reflection creates complex hyperbolic phase fringes.

Periodic lattice structures create absolute acoustic mirrors

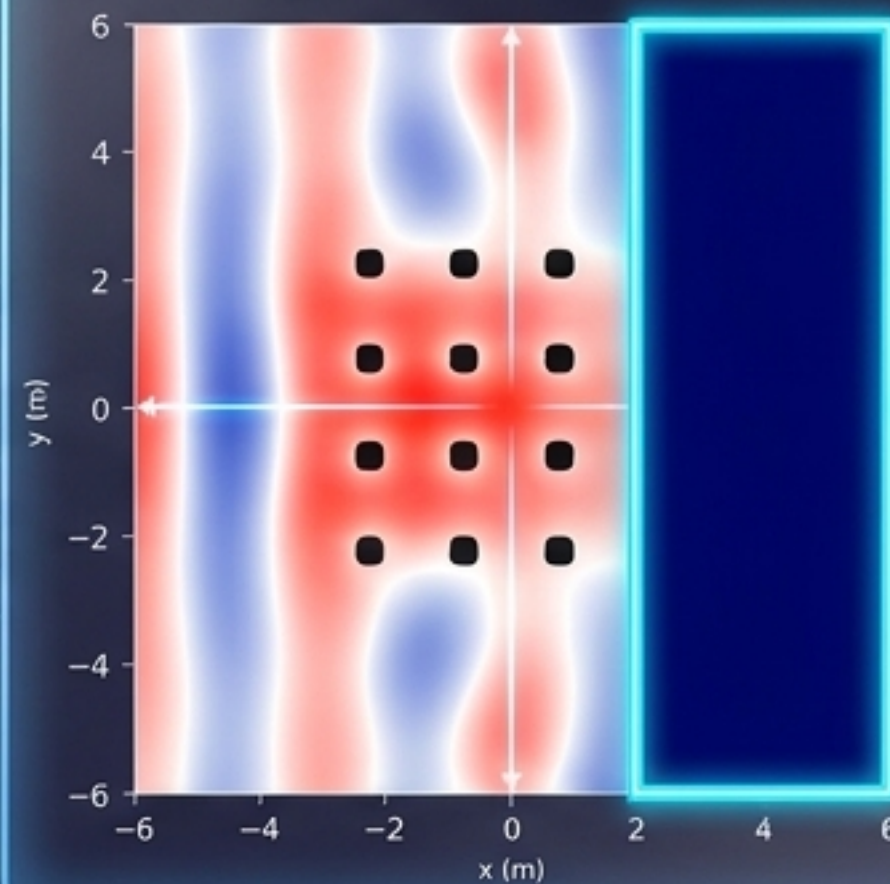
The Bragg Band Gap: When wavenumber aligns with lattice spacing ($k = \pi/d$), every row reflects in phase.
The Result: The entirely porous crystal transforms into an opaque wall, completely blocking transmission without requiring solid bulk mass.

1. Below Gap ($k=1.0$)



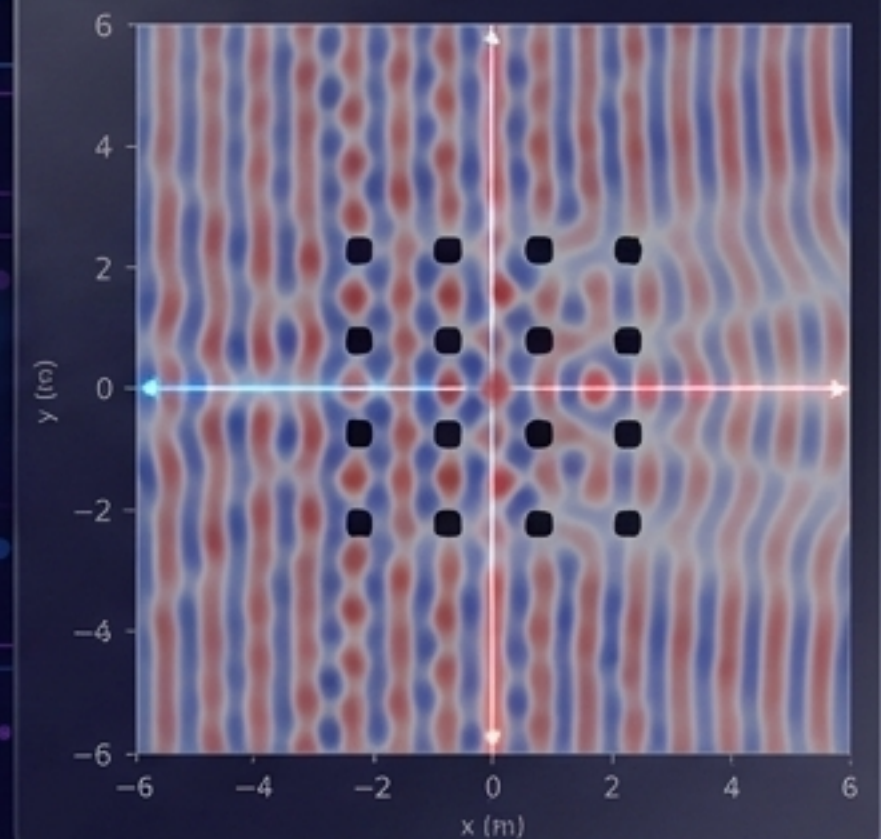
$k=1.0$

2. Bragg Condition ($k=2.09$)



$k=2.09$

3. High Freq ($k=8.0$)

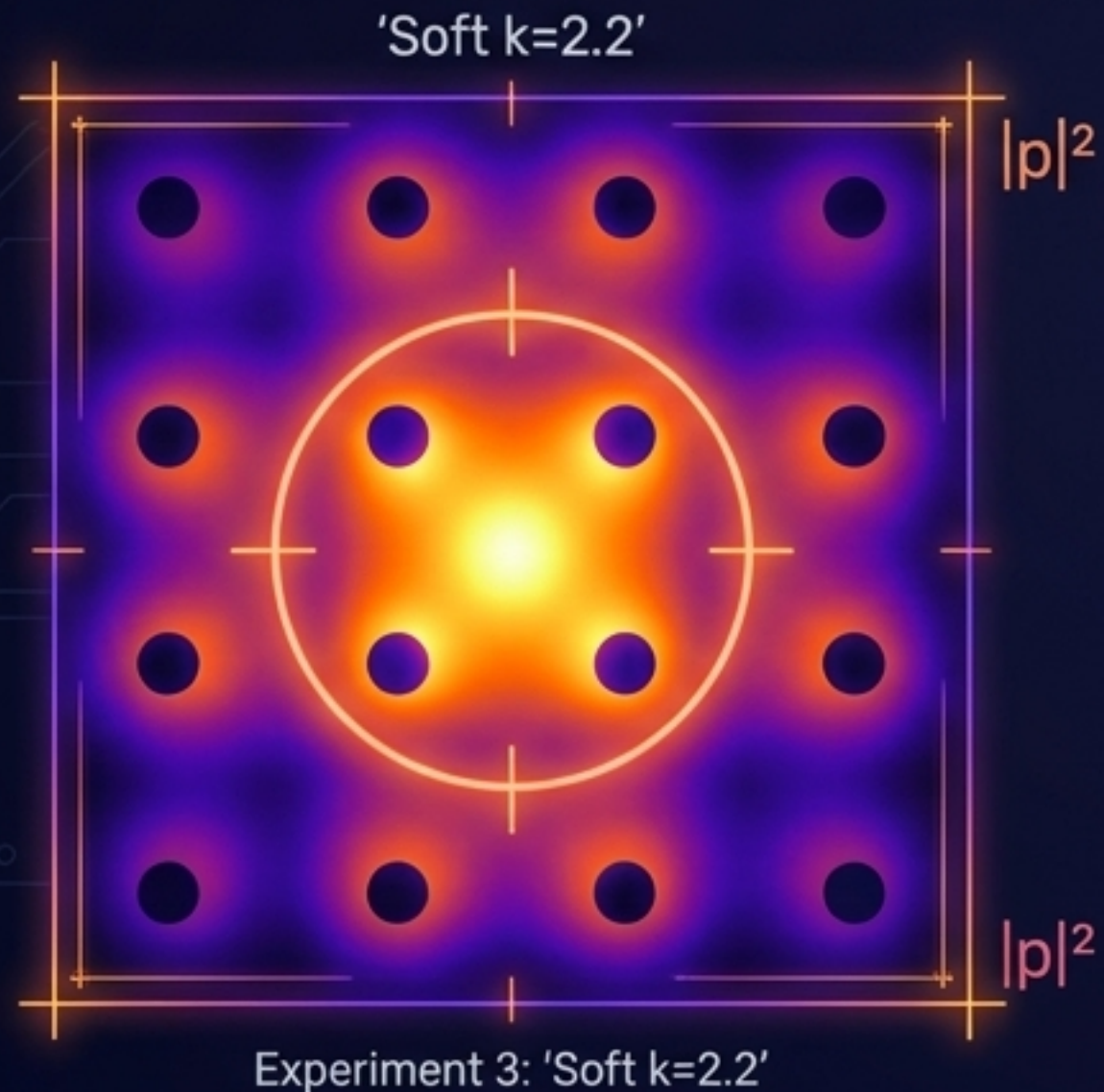


$k=8.0$

Manipulating the lattice traps and localizes acoustic energy

Trapped Cavity Resonance

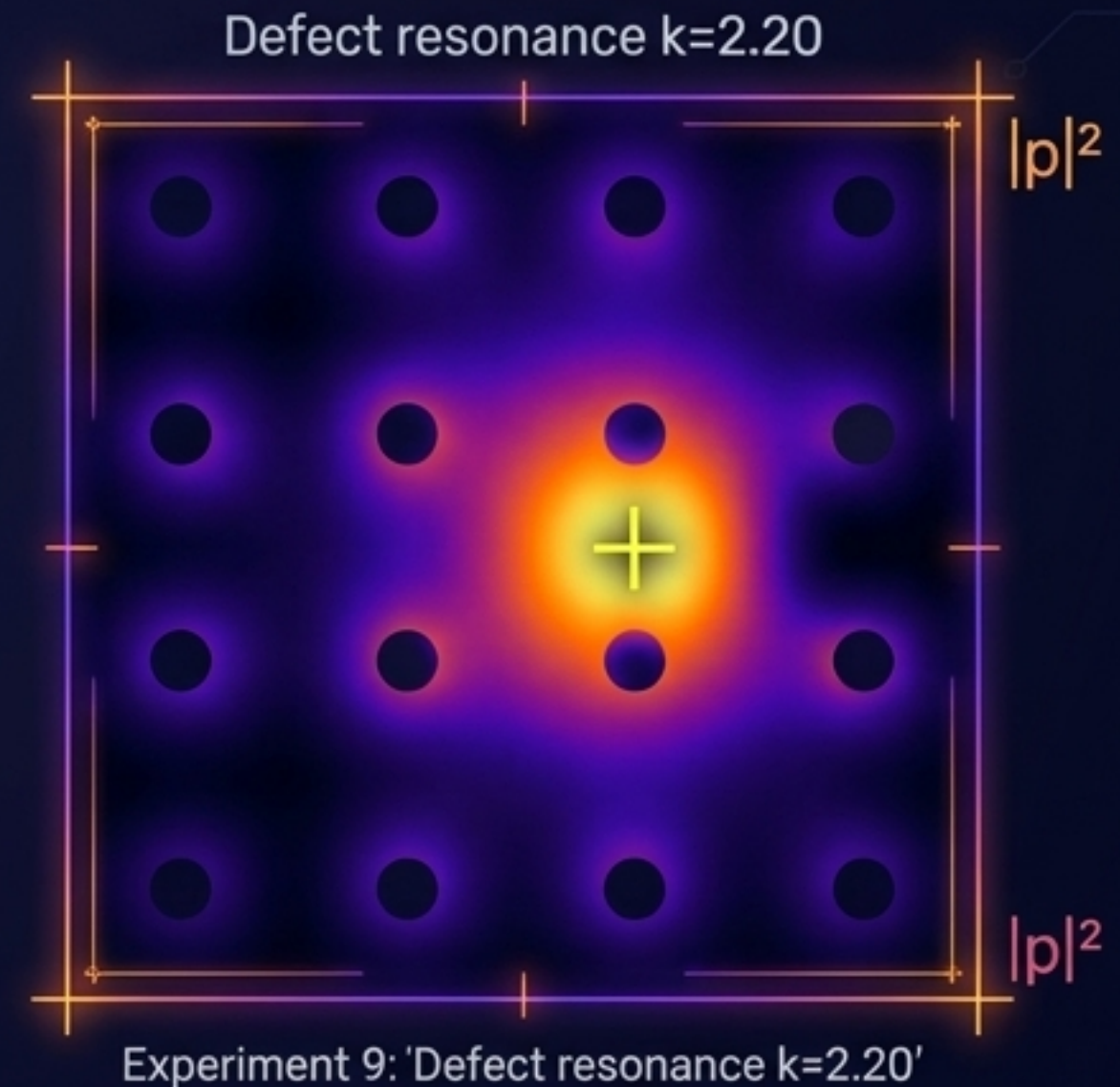
Operating just above the band gap locks energy inside the lattice via constructive interference, acting like an optical Fabry-Pérot cavity.



vs.

Defects as Features

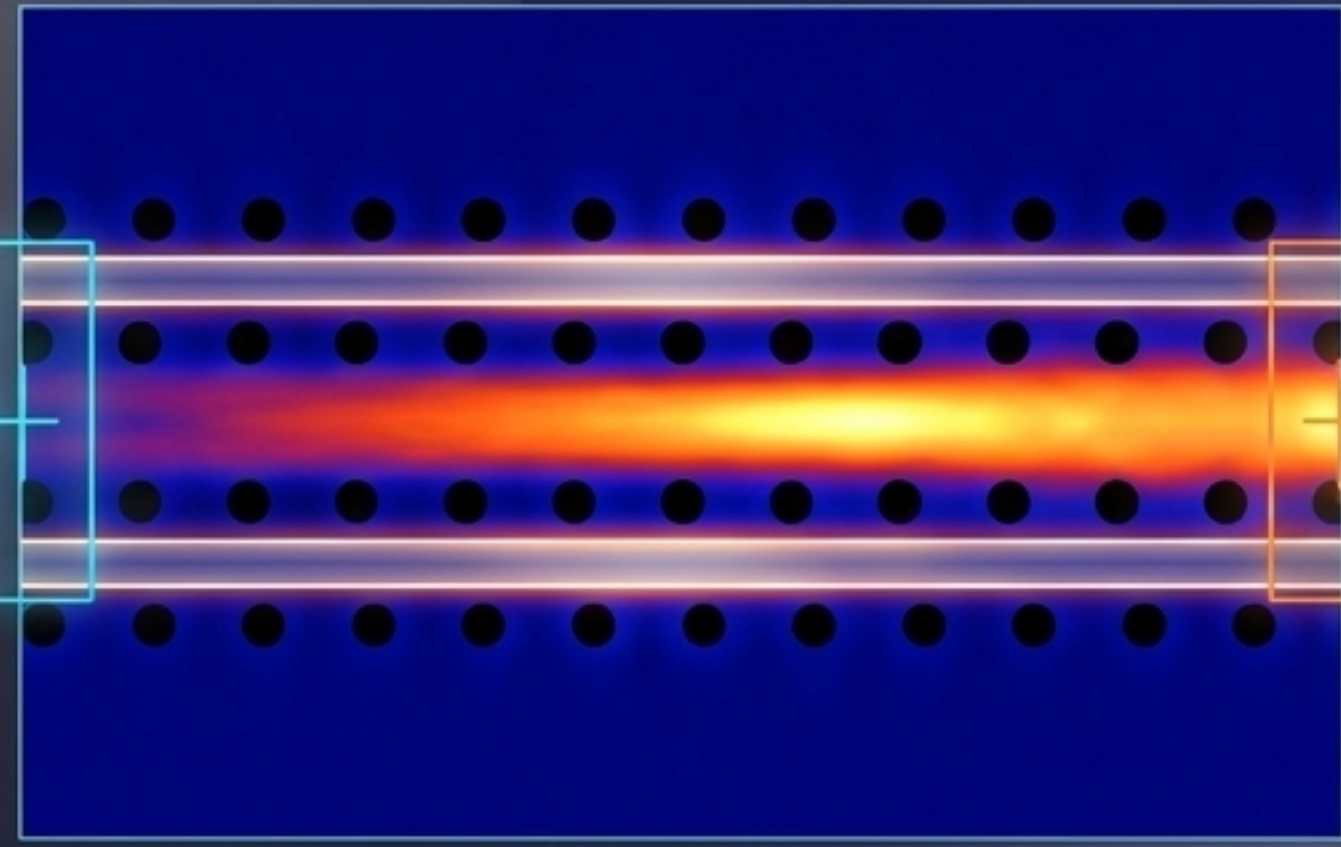
Removing a single rod creates a spatially localized hot spot inside the band gap. Energy cannot propagate through the surrounding crystal, so it pools precisely at the defect.



Acoustic energy can be structurally steered and routed

Linear Confinement

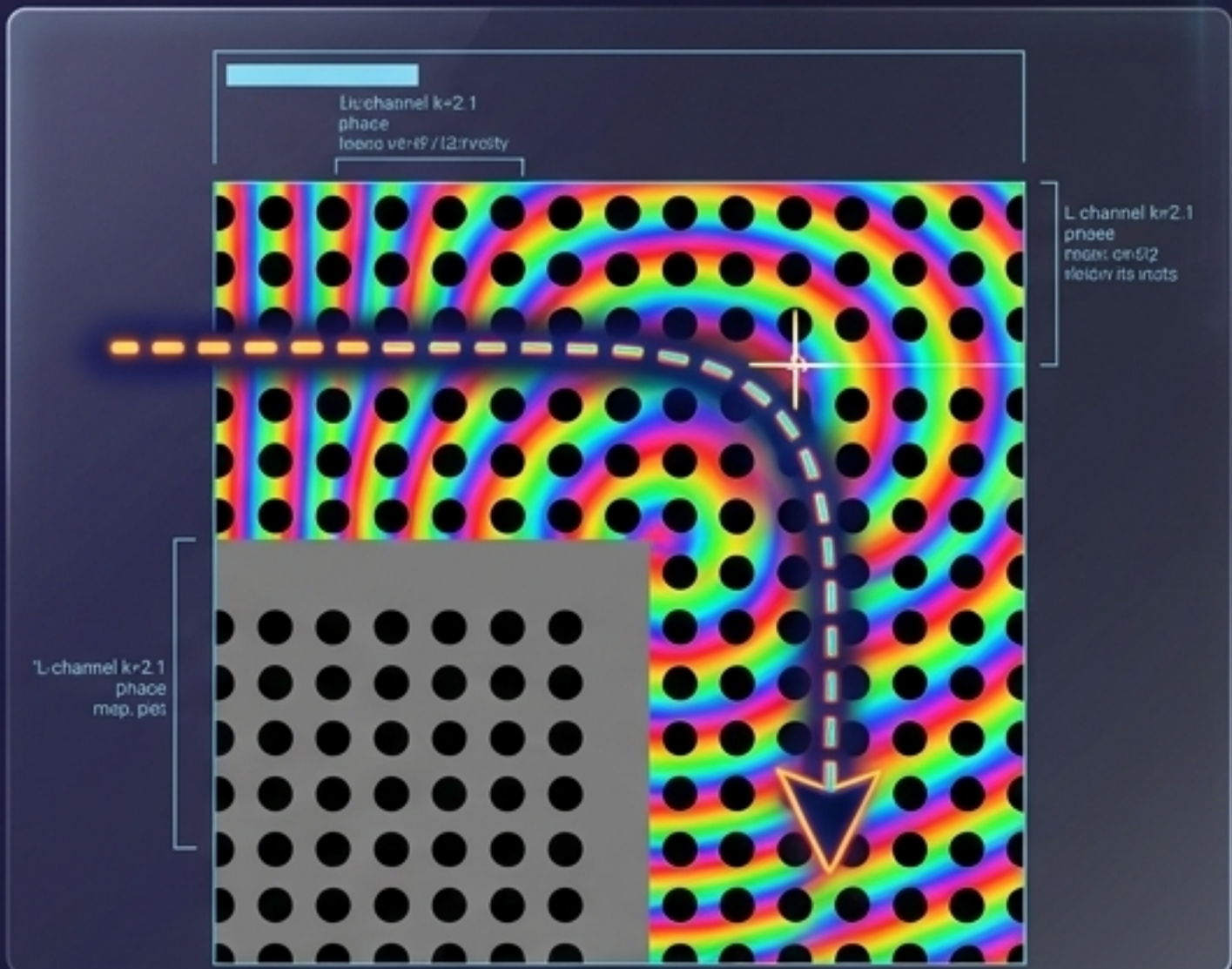
Rigid boundaries spaced correctly ($k \cdot w/2 \approx \pi/2$) funnel energy into a tightly collimated beam rather than allowing it to spread.



'Hard $k=2.2$ $|p|^2$ intensity'

Topological Routing

Carving an L-shaped path out of a sonic crystal forces waves around a sharp 90-degree bend. The surrounding band gap forbids leakage, ensuring perfect transmission.

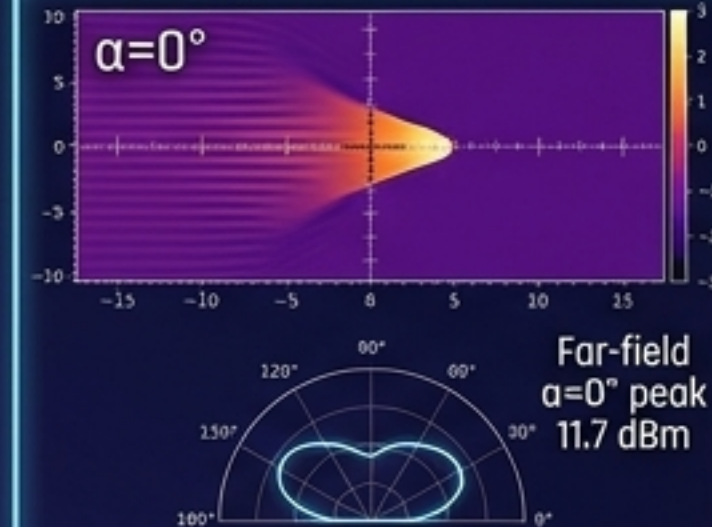


'L-channel $k=2.1$ phase'

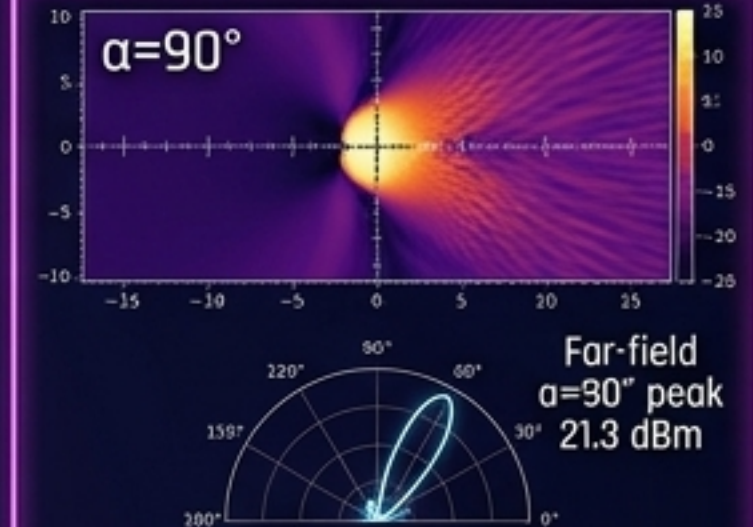
Geometric shape dynamically dictates scattering behavior

Asymmetric Stealth

Conformal geometries like the Joukowski profile exploit asymmetry. A sharp trailing edge yields acoustic stealth, while the blunt leading edge scatters massively.

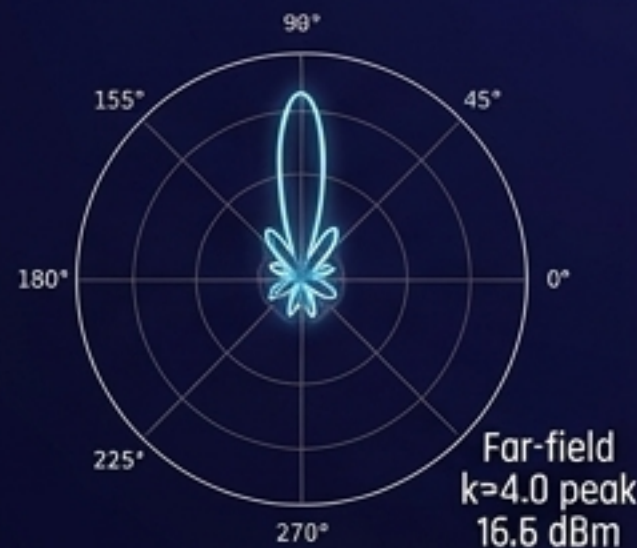
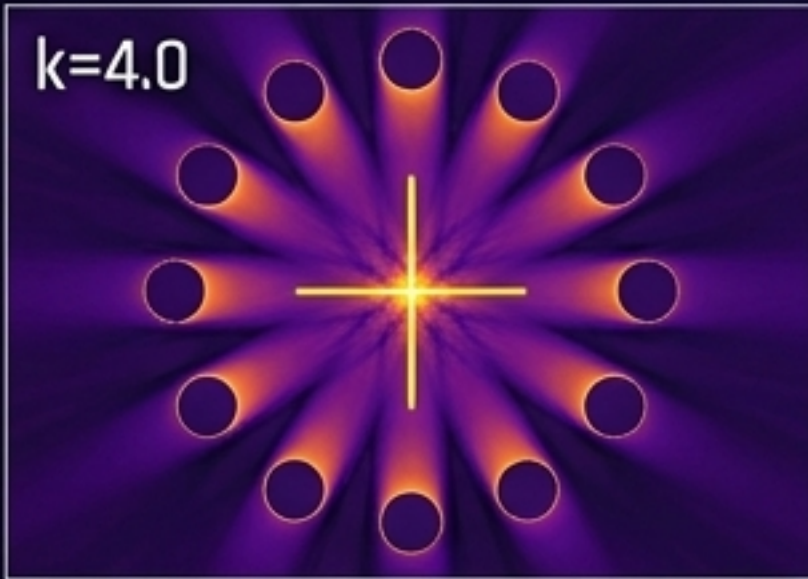


Exp 7 — Joukowski wing: field maps (top) and far-field patterns (bottom) $k=8$



Exp 7 — Joukowski wing: field maps (top) and far-field patterns (bottom) $k=8$

$k=4.0$



Exp 8 — Parabolic reflector: intensity (top) + far-field (bottom)

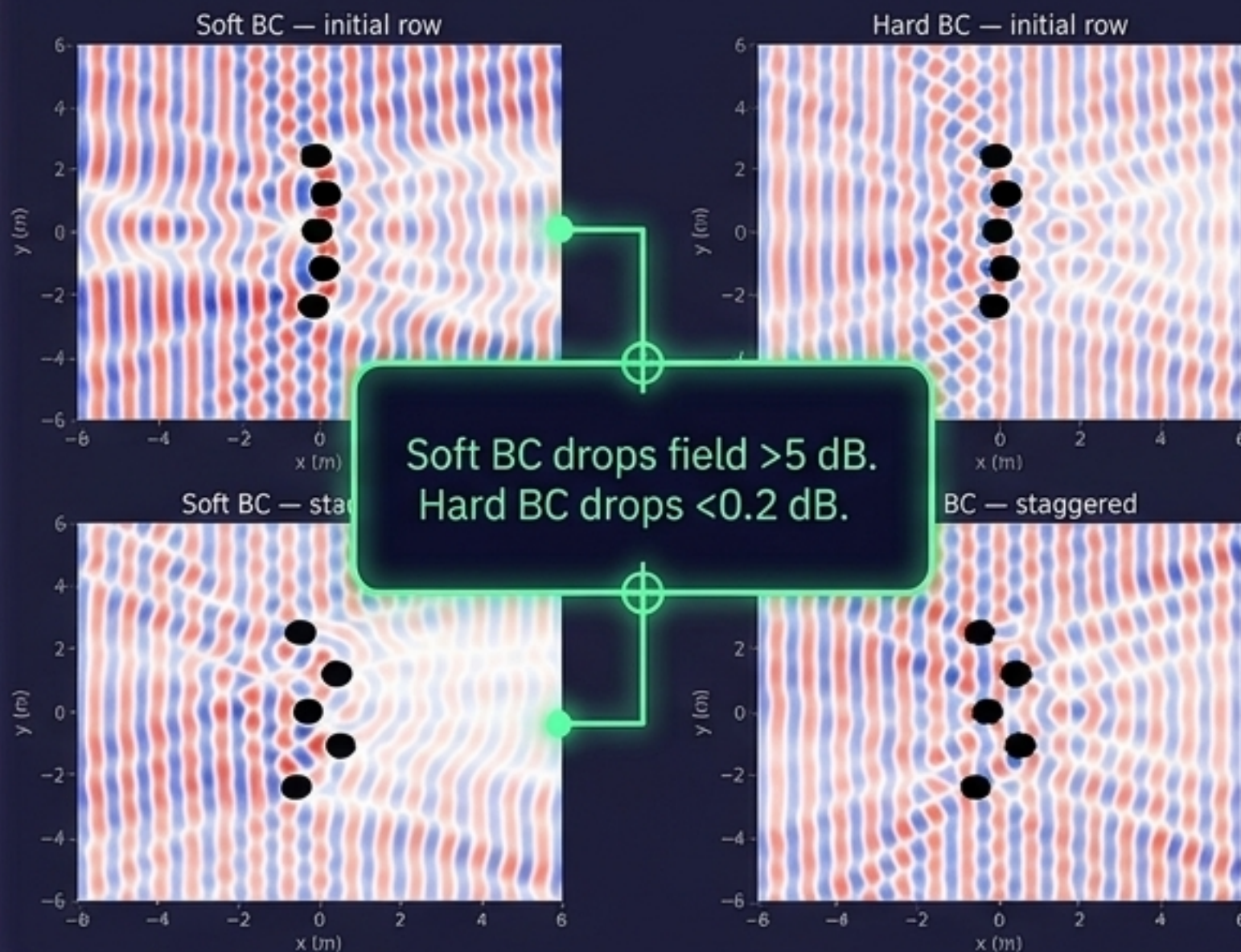
Geometric Focusing

A parabolic arrangement of rigid obstacles acts as a lens, converging plane waves into a tight, high-intensity focal point (analogous to a lithotripter).

Real-world acoustic architecture relies on material and diffusion

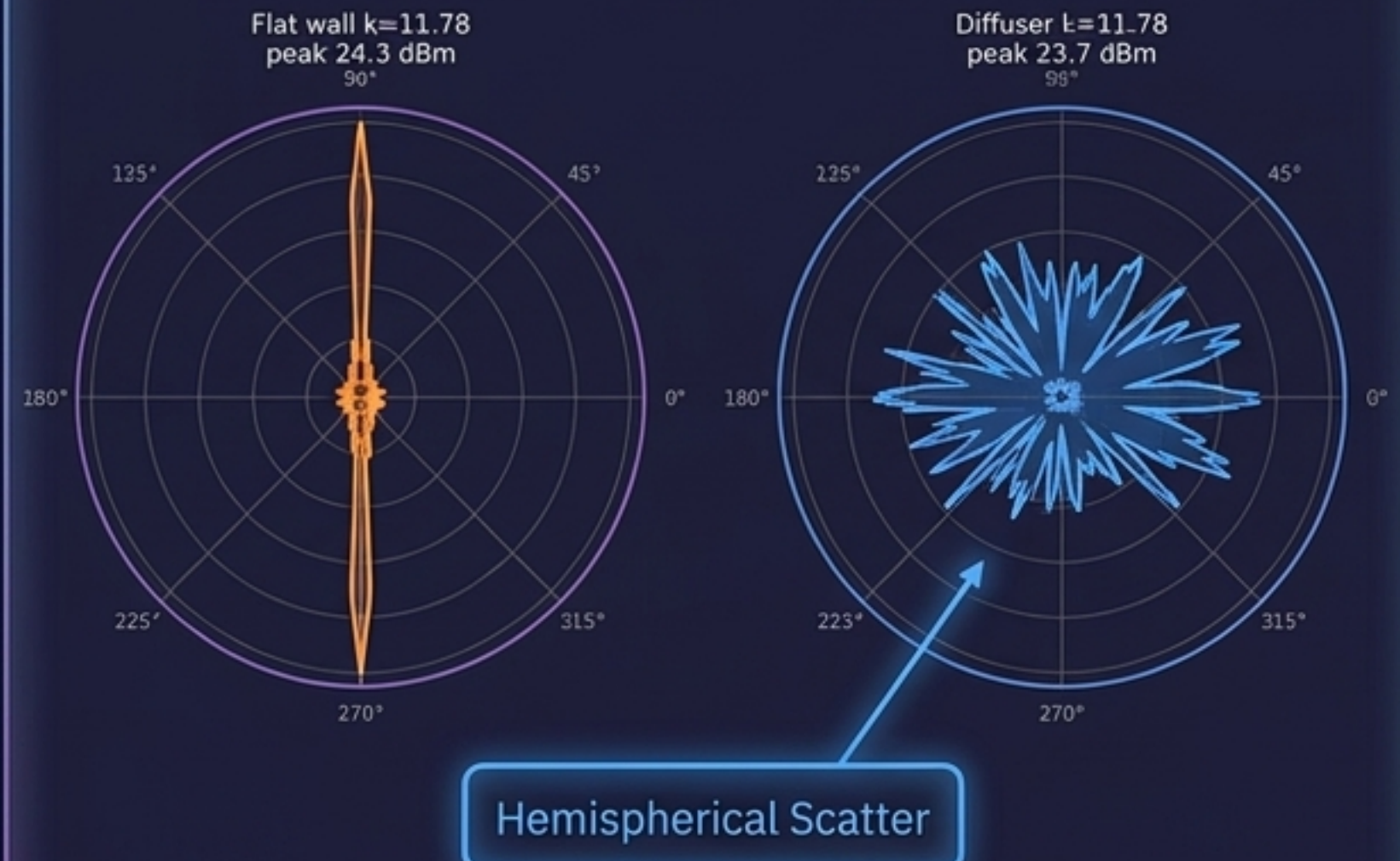
Noise Barriers (Exp 6)

Material > Arrangement: Algorithmic optimization proves that dense, sound-soft materials (vegetation) absorb energy deeply. Re-arranging rigid materials (steel poles) merely redirects noise without killing it.



Diffusers (Exp 11)

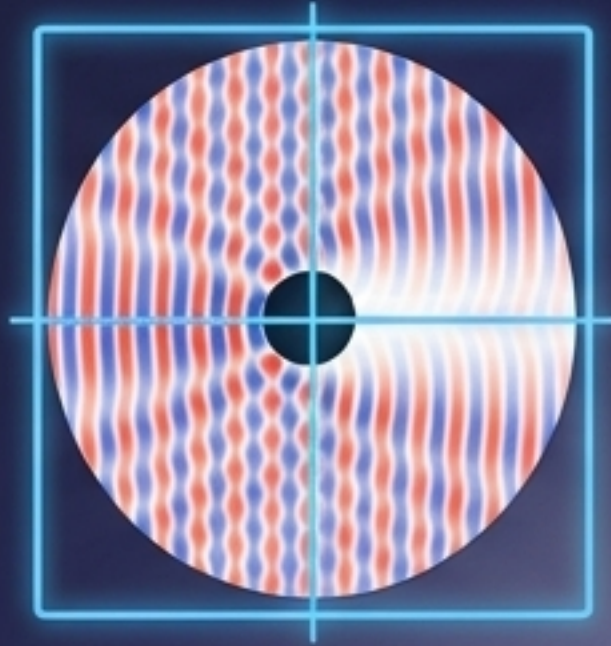
Diffusion > Reflection: A Schroeder diffuser uses a quadratic residue sequence of varying depths to shatter a harsh specular echo into a smooth, hemispherical envelope.



The 4 fundamental levers of acoustic design

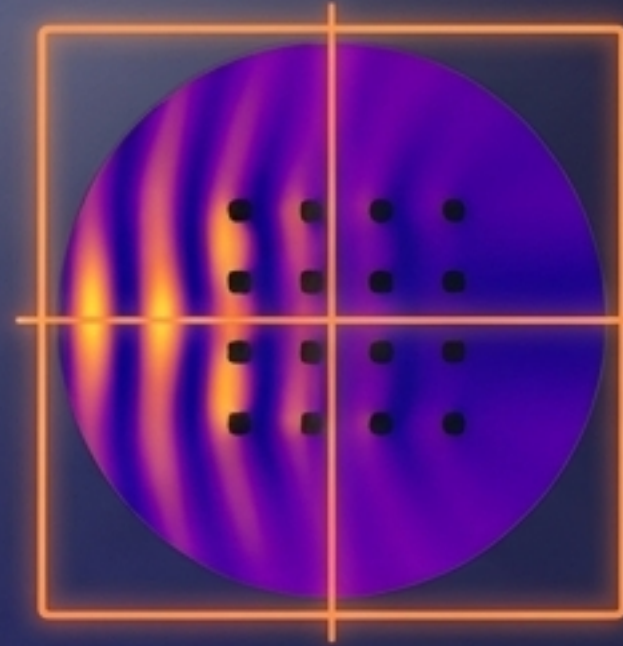
1. Material First

Absorption (Soft BC) mathematically outperforms rigid reflection for genuinely removing energy from a system.



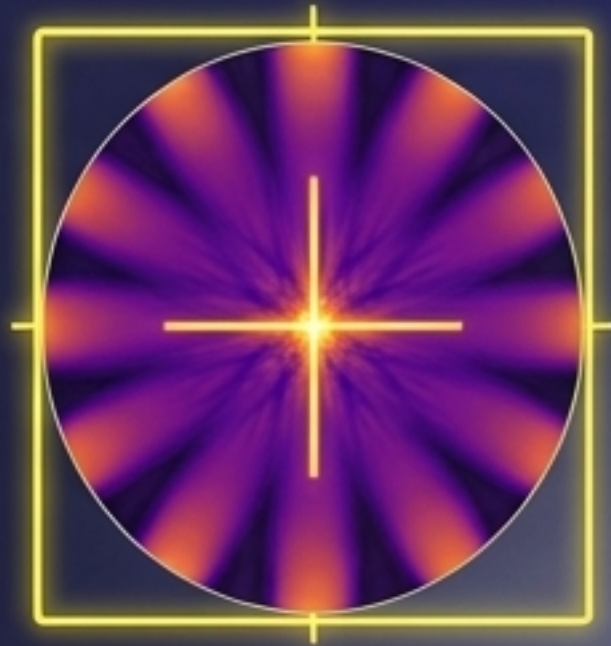
2. Embrace Periodicity

Bragg physics and structural band gaps can block or trap waves far more efficiently than sheer bulk mass.



3. Shape Directs Energy

Scatterer geometry directly dictates the outcome—whether evading detection entirely or focusing power to a pinpoint.



4. Diffuse, Don't Reflect

When hard boundaries are unavoidable, introducing structured depth variation randomizes phase and eliminates flutter echoes.

